Exam 2025 solutions

B—bookwork, E—new example, A—new application, T—new theory

1.

a)

[2E]

$$Z = X + Y > 0, \qquad W = X - Y > 0$$

$$x_1 = \frac{z+w}{2}, \qquad y_1 = \frac{z-w}{2}$$

[3E]

is the only solution. Moreover

$$J = \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = -2$$

so that

[2E]

$$f_{ZW}(z,w) = \frac{1}{|J|} f_{XY}(x_1, y_1) = \frac{1}{2}\, e^{-(z+w)/2}, \quad 0 < w < z < \infty$$

[3E]

$$\begin{aligned} F_Z(z) &= \int_0^z f_{ZW}(z,w)\, dw = \tfrac{1}{2}\, e^{-z/2}\ \frac{e^{-w/2}}{(-1/2)}\Big|_0^z \\ &= \tfrac{1}{2}\, e^{-z/2}\ \frac{e^{-w/2}}{(-1/2)}\Big|_0^z = e^{-z/2}(1 - e^{-z/2}), \quad z > 0 \end{aligned}$$

b) Step 1: Find *k* using normalization

[2B]

$$\int_0^2 kx^2 dx = 1.$$

Then

[3B]

$$\frac{kx^3}{3}\Big|_0^2 = 1 \Rightarrow \frac{k(8)}{3} = 1 \Rightarrow k = \frac{3}{8}.$$

Step 2: Compute $E[X]$:

[2B]

$$E[X] = \int_0^2 x f_X(x) dx = \int_0^2 x \cdot \frac{3}{8} x^2 dx.$$

[3B]

$$E[X] = \frac{3}{8}\int_0^2 x^3 dx = \frac{3}{8} \times \frac{16}{4} = \frac{3}{8} \times 4 = \frac{12}{8} = 1.5.$$

c) Possible pairs: (1,6),(2,5),(3,4),(4,3),(5,2),(6,1). [2E]

There are 6 favorable outcomes out of 36 total (since 6×6=36). [1E]

$$P(\text{sum} = 7) = \frac{6}{36} = \frac{1}{6}.$$

[2E]

2.

a) The joint distribution of the samples is given by

$$f_X(x_1, \dots, x_n; \lambda) = \prod_{i=1}^{n} e^{-\lambda} \frac{\lambda^{x_i}}{x_i!} = e^{-n\lambda} \frac{\lambda^{\sum x_i}}{\prod x_i!} \quad \text{[2E]}$$

Then the log-likelihood function is [3E]

$$\ln f_X(x_1, \dots, x_n; \lambda) = -n\lambda + \sum x_i \ln\lambda - \ln\prod x_i!$$

Now take the derivatives:

$$\frac{\partial}{\partial\lambda} \ln f_X(x_1, \dots, x_n; \lambda) = -n + \frac{\sum x_i}{\lambda}$$

$$\frac{\partial^2}{\partial\lambda^2} \ln f_X(x_1, \dots, x_n; \lambda) = -\frac{\sum x_i}{\lambda^2} \quad \text{[3E]}$$

Then the Fisher information is given by

$$I = -E\left[-\frac{\sum x_i}{\lambda^2}\right] = \frac{n\lambda}{\lambda^2} = \frac{n}{\lambda} \quad \text{[2E]}$$

b) So the Cramer-Rao bound is

$$\mathrm{Var}[\hat{\lambda}] \geq \frac{1}{I(\lambda)} = \frac{\lambda}{n} \quad \text{[5E]}$$

c)

i) The mean is given by

$$E[\hat{\theta}] = E[\frac{1}{n}\sum_{i=1}^{n} X_i] = \frac{1}{n}\sum_{i=1}^{n} \lambda = \lambda \quad \text{[3E]}$$

So it is unbiased. [1E]

ii) Since X_i is Poisson, its variance is

$$\mathrm{Var}(X_i) = \lambda$$

Since X_i 's are i.i.d., the variance of the sample mean is:

$$Var(\hat{\lambda}) = \frac{1}{n} Var(X_i) = \frac{\lambda}{n} \quad \text{[3E]}$$

[1E]

which tends to 0, so it is consistent.

iii) It also achieve the CRLB, hence an efficient estimator. [2E]

3.

a)

i) matched filter

$$h(t) = s(t_0 - t) = \begin{cases} \frac{1}{2}\cos(\pi(1-t)) = -\frac{1}{2}\cos(\pi t), & 0 < t < 1 \\ 0, & \text{otherwise} \end{cases}$$

[5E]

ii) maximum SNR

$$SNR_{max} = \frac{E_s}{N_0} = \frac{\int_0^1 s(t)^2 dt}{N_0} = \int_0^1 s(t)^2 dt$$

$$= \frac{1}{4}\int_0^1 \cos(\pi t)^2\, dt = \frac{1}{4}\int_0^1 \frac{1+\cos(2\pi t)}{2} dt = \frac{1}{8}$$

[5E]

b)

i) The autocorrelation function is

$$R_X(k) = \begin{cases} 1+\alpha^2 & if\ k = 0 \\ \alpha & if |k| = 1 \\ 0 & \text{otherwise} \end{cases}$$

[3E]

Hence the power spectral density is

$$S_X(\omega) = R_X(0) + R_X(1)e^{-j\omega} + R_X(-1)e^{j\omega} = 1 + \alpha^2 + 2\alpha\cos(\omega), \quad |\omega| < \pi$$

[3E]

ii) In the general case, the autocorrelation function is

[3E]

$$R_X(k) = \sum_{i=0}^{r} \alpha_i \alpha_{k+i}$$

where the convention that $\alpha_s = 0$ if $s < 0$ or $s > r$. Thus,

[3E]

$$S_X(\omega) = \sum_{k=-\infty}^{\infty} R_X(k)e^{-jk\omega} = \sum_{k=-\infty}^{\infty} \sum_{i=0}^{r} \alpha_i \alpha_{k+i}\, e^{-jk\omega}$$

$$= \sum_{k=-\infty}^{\infty} \alpha_{k+i} e^{-j(k+i)\omega} \sum_{i=0}^{r} \alpha_i e^{ji\omega}$$

$$= \sum_{k'=0}^{r} \alpha_{k'} e^{-j(k')\omega} \sum_{i=0}^{r} \alpha_i e^{ji\omega} = |A(e^{j\omega})|^2, \quad |\omega| < \pi$$

where $A(Z) = \sum_{i=0}^{r} \alpha_i Z^i$ (the last step holds because that $\alpha_s = 0$ if $s < 0$ or $s > r$).

4.

a) i) From the equation $\pi = \pi P$, we obtain

$$\pi_A = 0.4\pi_A + 0.2\pi_B + 0.3\pi_C$$
$$\pi_B = 0.5\pi_A + 0.3\pi_B + 0.3\pi_C$$
$$\pi_C = 0.1\pi_A + 0.5\pi_B + 0.4\pi_C$$

[2E]

Also, we know the normalization condition:

$$\pi_A + \pi_B + \pi_C = 1$$

The answer is

[3E]

$$\pi_A = 0.32, \quad \pi_B = 0.28, \quad \pi_C = 0.4.$$

ii) To find $P_{AB}^{(2)}$, we compute $P^2 = P \times P$:

$$P^2 = \begin{bmatrix} 0.4 & 0.5 & 0.1 \\ 0.2 & 0.3 & 0.5 \\ 0.3 & 0.3 & 0.4 \end{bmatrix} \times \begin{bmatrix} 0.4 & 0.5 & 0.1 \\ 0.2 & 0.3 & 0.5 \\ 0.3 & 0.3 & 0.4 \end{bmatrix}$$

[2E]

$$= \begin{bmatrix} 0.29 & 0.38 & 0.33 \\ 0.29 & 0.34 & 0.37 \\ 0.30 & 0.36 & 0.34 \end{bmatrix}$$

[2E]

The probability of transitioning from state A to state B in two steps is:

[1E]

$$P_{AB}^{(2)} = 0.38$$

b) Denote the intensity by α (=0.2) and β (=0.25), respectively.

i) Let S (respectively T) denote the arrival time of the first white mouse (respectively black mouse). They have independent exponential distributions:

[2A]

$$f(s) = \alpha e^{-\alpha s}, \quad s > 0$$
$$f(t) = \beta e^{-\beta t}, \quad t > 0$$

That the first mouse is black means

$$T < S$$

The probability is given by

$$P(T < S) = \iint_{t<s} \alpha e^{-\alpha s} \beta e^{-\beta t} ds dt$$
$$= \int_{t=0}^{\infty} \beta e^{-\beta t} \int_{s=t}^{\infty} \alpha e^{-\alpha s} \, dt ds$$
$$= \int_{t=0}^{\infty} \beta e^{-\beta t} e^{-\alpha t} dt$$
$$= \frac{\beta}{\alpha + \beta} = \frac{5}{9}$$

[3A]

ii) Remember the Poisson process is memoryless, so

[2A]

$$P(T_1 < S_1, T_2 > S_2) = \frac{\beta}{\alpha + \beta} \frac{\alpha}{\alpha + \beta}$$
$$= \frac{\alpha\beta}{(\alpha + \beta)^2}$$
$$= \frac{20}{81}$$

[3A]

iii) Firstly, the total number of mice is a Poisson process with intensity 0.45.

[1A]

It can be thought of random selection of a Poisson process, which gives rise to two Poisson processes, each with arrival rate $\lambda = 0.225$. This yields two queues with different service times. The number of mice is therefore the sum of the two queues, i.e., $k = k_1 + k_2$, where k_1 is Poisson distributed with parameter 0.225x10 = 2.25 and k_2 is Poisson distributed with parameter = 0.225x20 = 4.5.

[2A]

We know that the sum of two Poisson distributions is again Poisson with parameter equal to the sum of the individual distributions' parameters, i.e., 6.75. So the average number of mice is 6.75.

[2A]